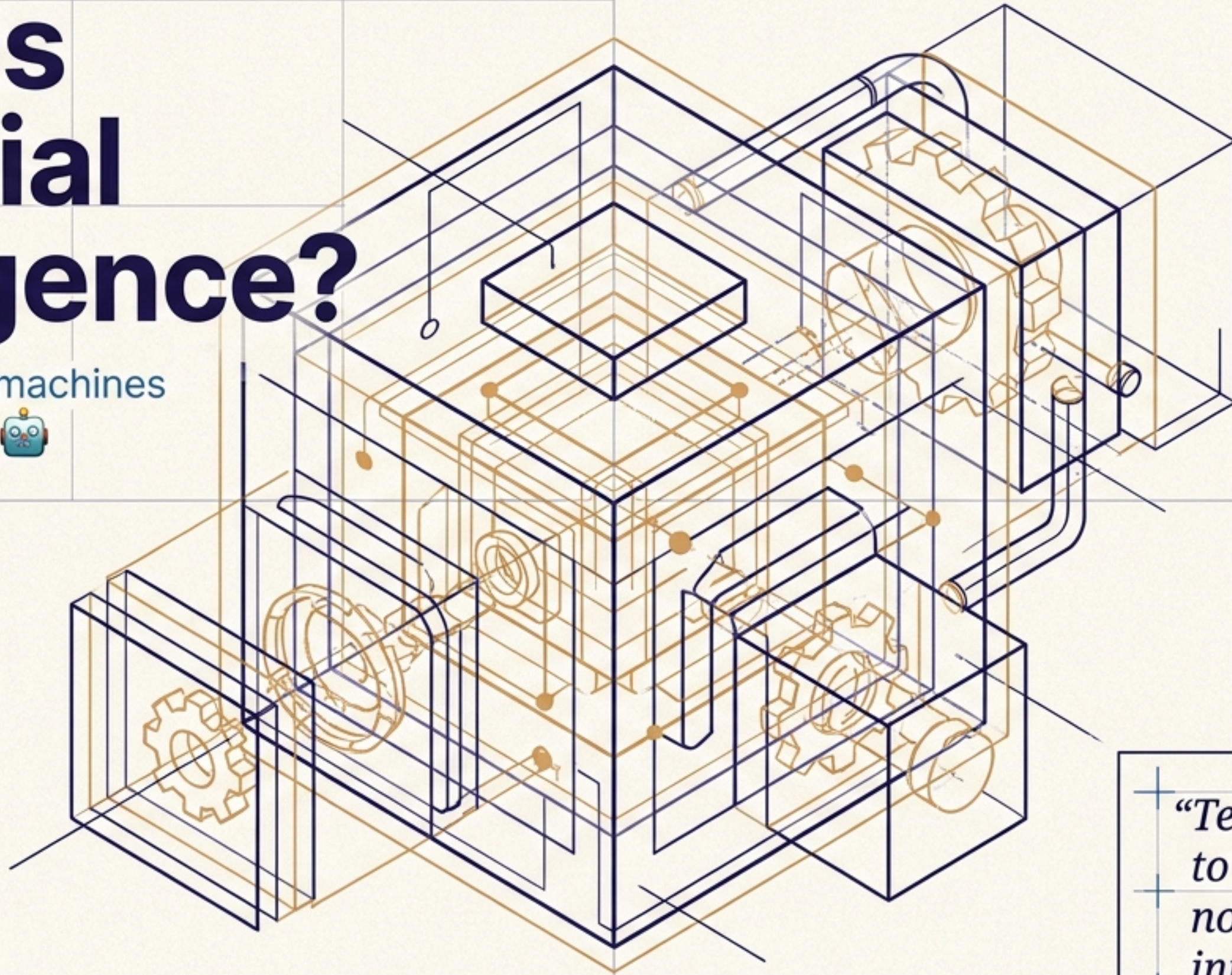


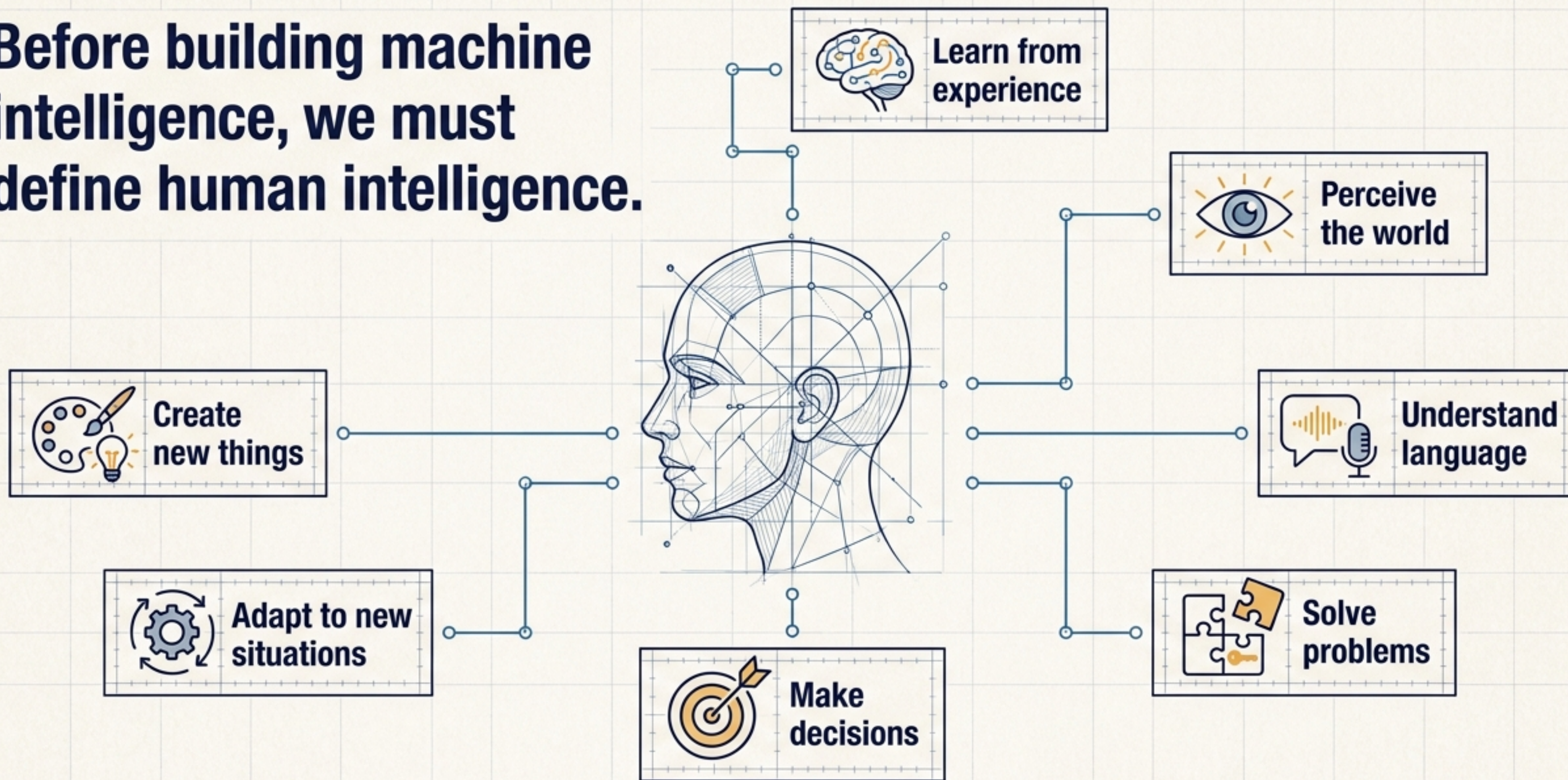
What Is Artificial Intelligence?

Understanding the machines behind the “magic” 🤖



“Teaching computers to do things that normally require intelligence.”

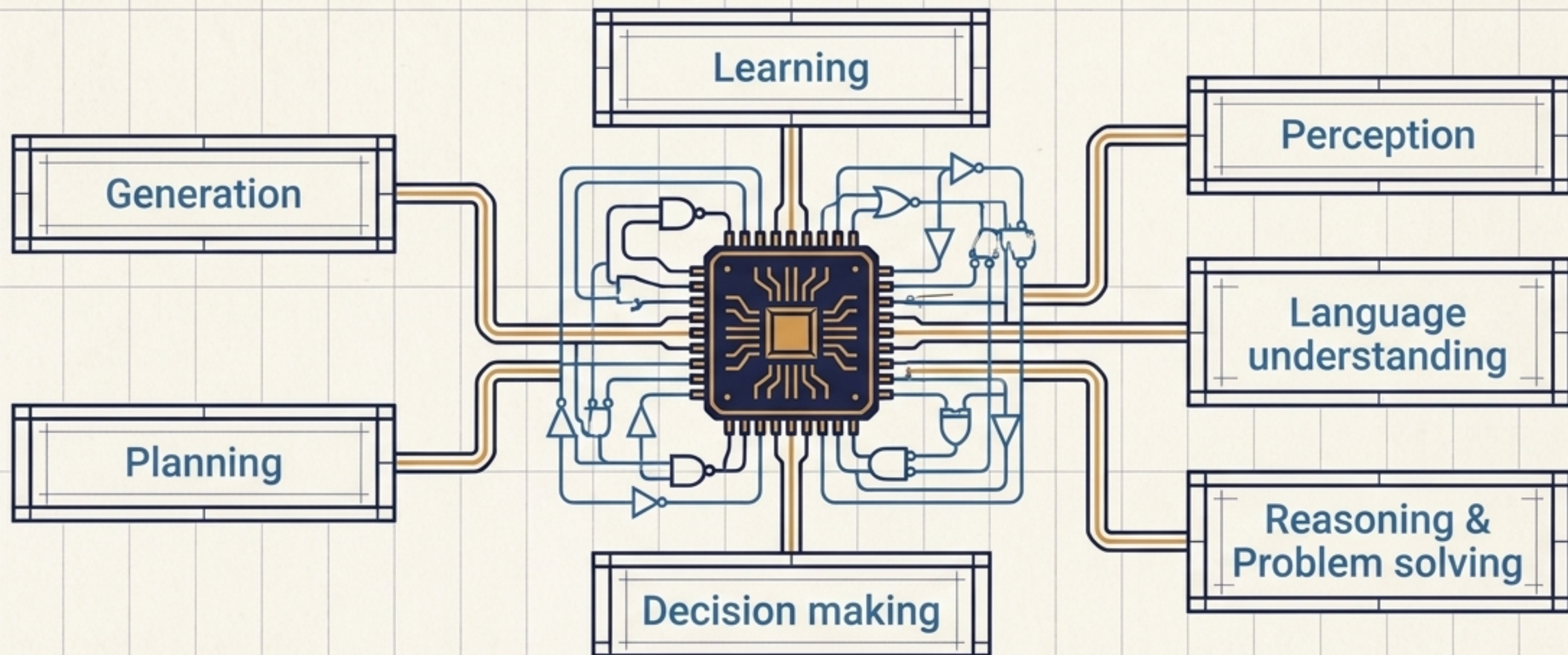
Before building machine intelligence, we must define human intelligence.



Can we make machines perform similar tasks? That's where Artificial Intelligence begins.

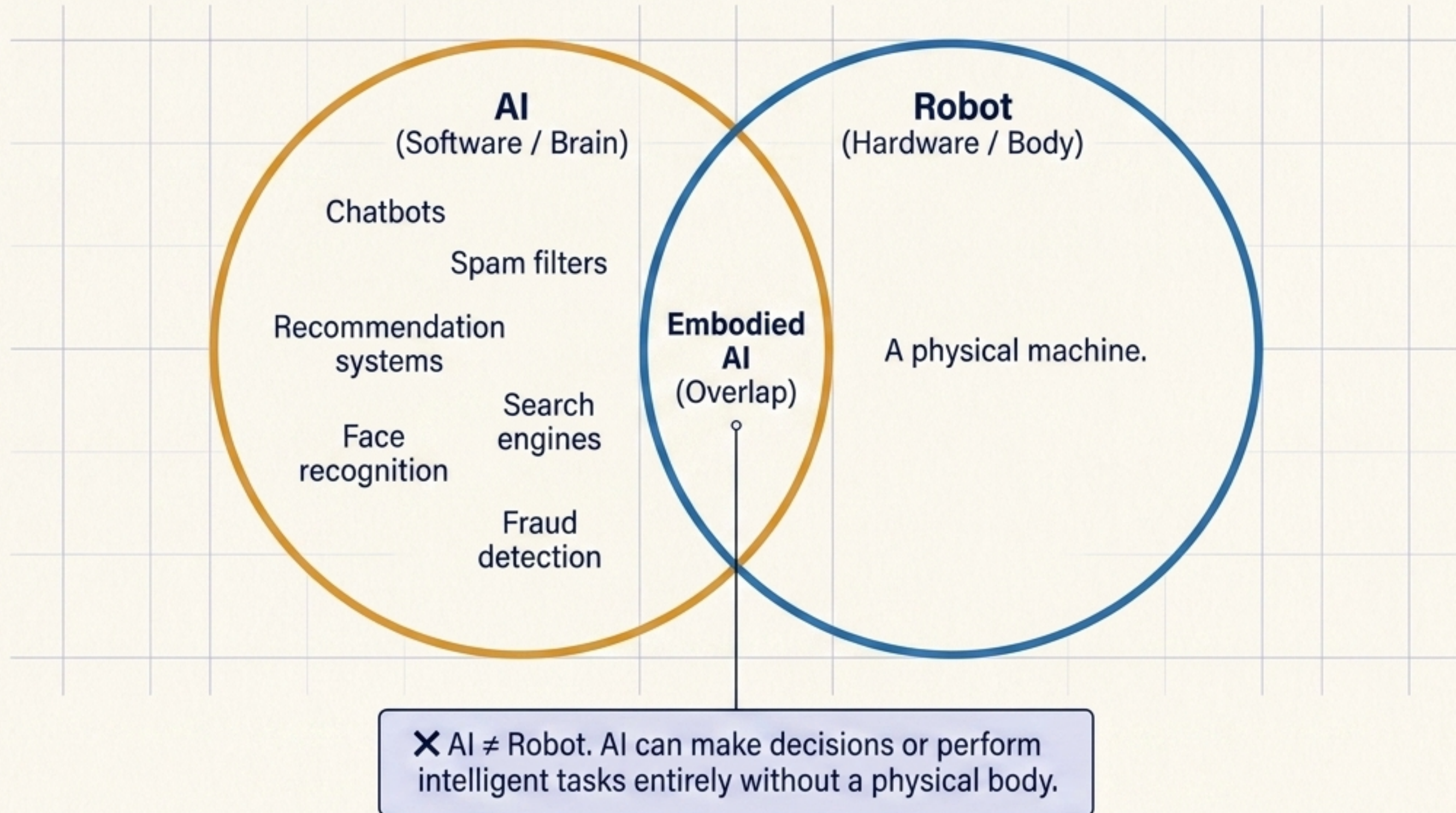
Artificial Intelligence maps biological cognitive tasks to digital systems.

AI is a field of computer science focused on creating systems capable of performing tasks that normally require human-like intelligence.



Simple version: AI = Making machines capable of intelligent behavior

Intelligence is the software, not the physical machine.



You were using AI long before you started studying it. 🦴

The Invisible Engine

The Application



Face Unlock



Spotify Recommendations



Spam Filter



Navigation



Product Recommendations



Chatbots



Photo Enhancement

The AI Engine

Computer Vision

Machine Learning

Classification

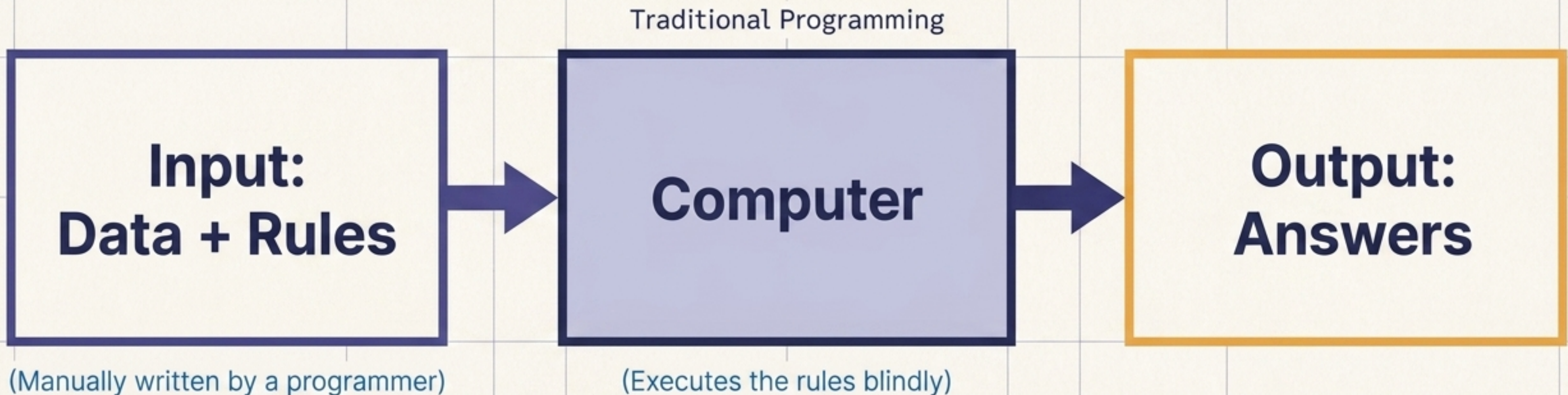
Prediction + Optimization

Machine Learning

Language Models

Deep Learning

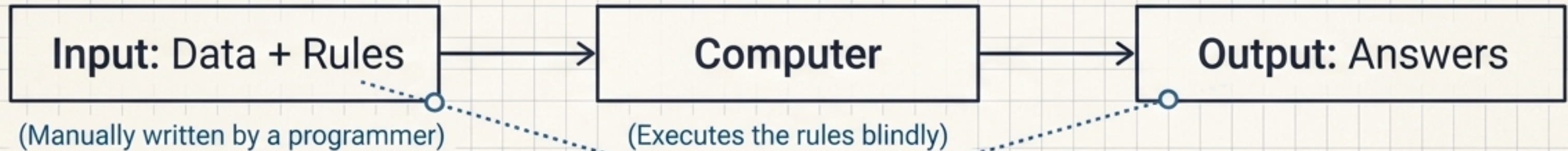
The legacy computing approach required humans to explicitly write every rule.



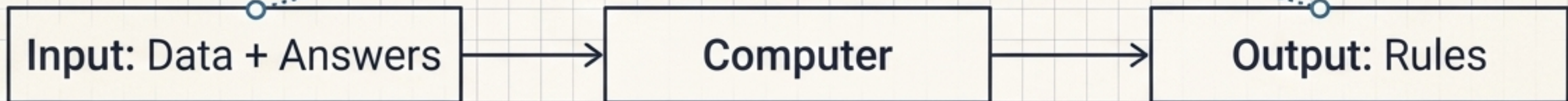
Before Machine Learning, the computer didn't discover the rules.
The programmer explicitly gave it the rules.

Machine Learning flips traditional computer science on its head.

Traditional Programming



Machine Learning



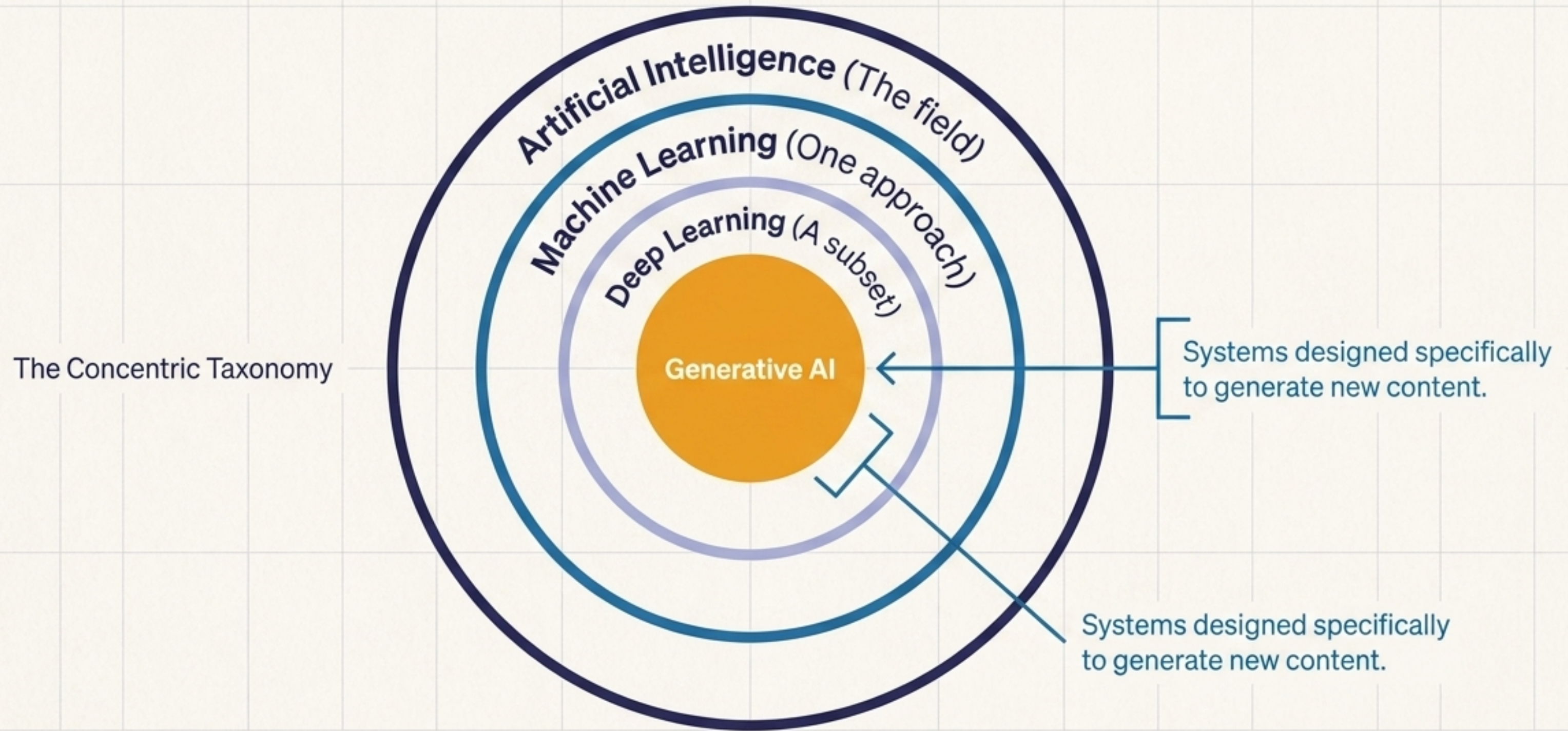
Instead of manually writing every rule, we give the system thousands of examples (like labeled emails). The model learns patterns from those examples to write its own rules.

The six modular capabilities of an artificial intelligence system.

The Capability Hex Grid

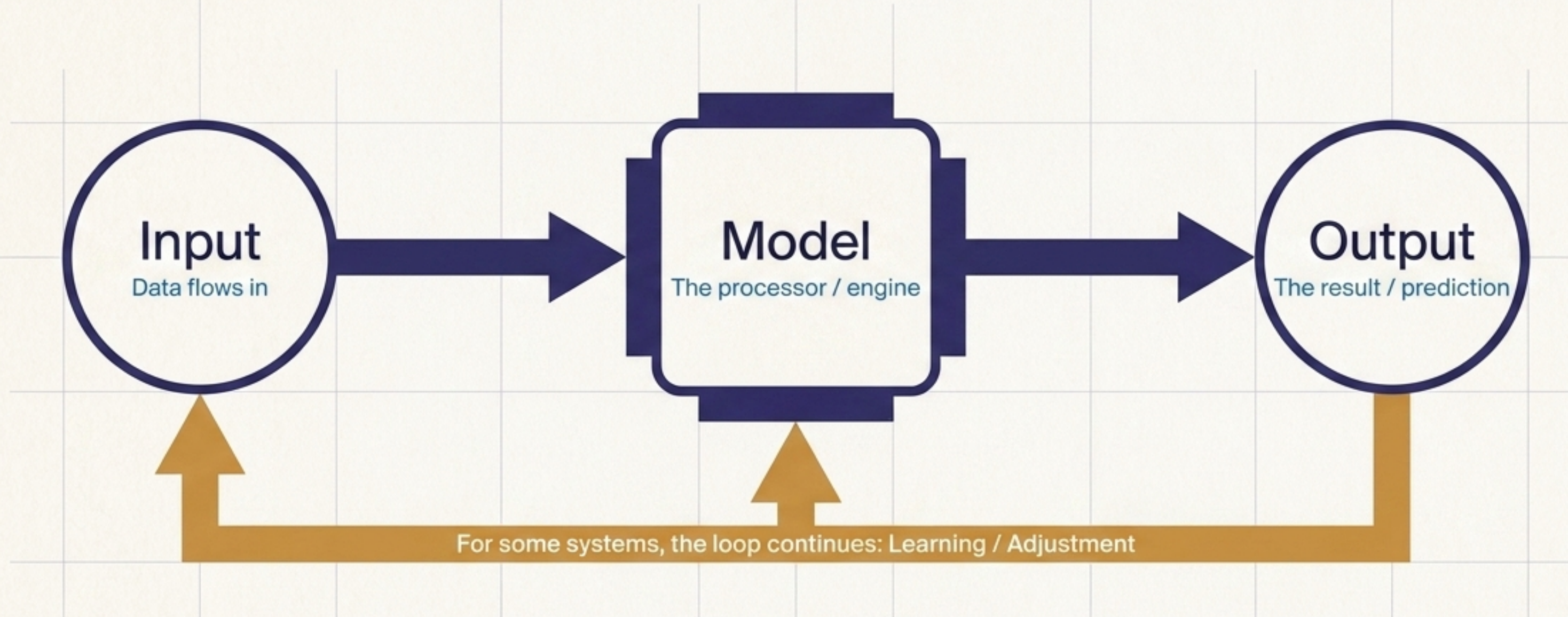


Mapping the categorical relationships within the AI ecosystem.



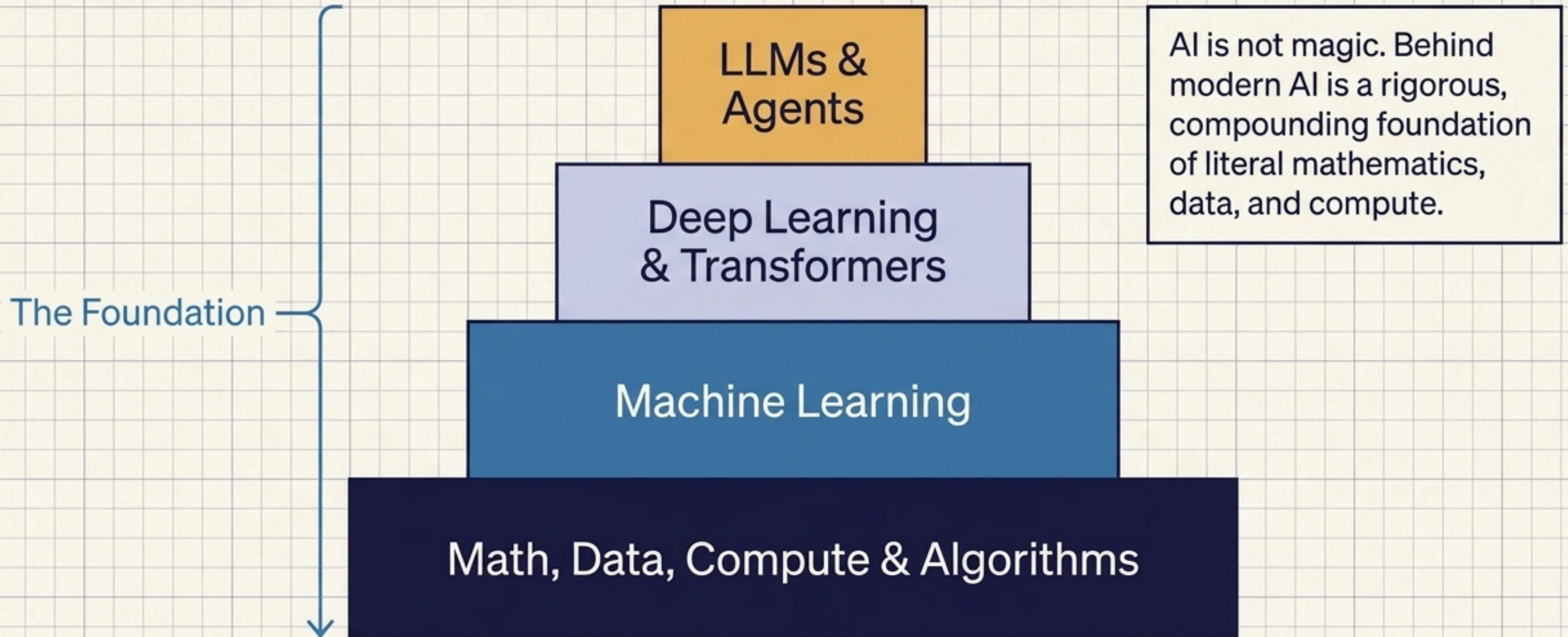
Remember: AI is the field. ML is one approach to AI. DL is a subset of ML.

The continuous mechanical loop behind how an AI system learns.



At a very high level, this is how it works.
Later we'll learn what actually happens inside each stage.

Strip away the 'magic' and you find a mathematical stack.



The Master Blueprint: 5 core truths to remember.

1	AI is a field, not a single technology.
2	AI is about creating systems capable of intelligent behavior.
3	AI does not require a physical robot.
4	Machine Learning is one major approach to building AI.
5	Modern AI is built on mathematics, data, algorithms, and computing.

The Journey Ahead.

"I know AI. I use ChatGPT."

"Maybe I don't know AI. 🦴"

**We are not just going to learn how to use AI.
We're going to learn how AI actually works.**